MODULE 16 · SECTION 2 - GOOGLE CLOUD'S GEN AI OFFERINGS

What an agent is: loops, types and teams of agents

How agents differ from bots and assistants, the loop they run, the parts inside them, and when not to use one.

SECTION 2 of 4 EXAM WEIGHT ~35% MODULE 16 of 33

admin@sthithapragna.com

A model answers. An agent **pursues a goal**: it reasons, plans and acts until the job is done.

admin@sthithapragna.com

Bot, assistant or agent?

KIND	HOW IT BEHAVES
Bot	Follows set rules; reacts to triggers
AI assistant	Responds and recommends; the user decides
AI agent	Goal-driven; decides and acts over many steps

THE TIEBREAKER

Who makes the decisions? The rules, the user, or the system itself. That tells you which one a question describes.

Observe, reason, plan, act, and check

01
Observe
Take in the request and the situation.

02
Reason
Work out what is going on.

03
Plan
Choose the steps and the tools.

04
Act
Call a tool to do a step.

05
Check
Look at the result; loop again.

ON THE EXAM

Reasoning and acting interleave until the goal is met. This is the ReAct pattern, which module 20 returns to.

The parts inside every agent

PART	ITS JOB
Model	The brain: understands, reasons, decides
Instructions	The role, the goal and the rules to follow
Tools	The hands: APIs, data and actions
Memory	What happened earlier, and what it learned
Orchestration	Runs the loop and keeps it on track

THE SIGNAL

A question that describes a capable agent that cannot do anything is missing tools. One that forgets is missing memory.

The returns agent, round the loop

STEP	WHAT THE RETURNS AGENT DOES
Observe	Reads the email and the photo of a split seam
Reason	Damaged in transit, and within the 30 days
Plan	Check the order, book a collection, confirm
Act	Calls the orders system, then the courier API
Check	Collection booked? Then it replies

THE TAKEAWAY

Each Act step uses a tool, and each result feeds the next Observe. That is what turns a chatbot into an agent.

Conversational or background?

CONVERSATIONAL

Talks with a person

- Started by someone's request
- Chat or voice, in real time
- Answers and completes tasks
- A shopping or support agent

BACKGROUND

Works behind the scenes

- Started by an event or schedule
- Little or no human conversation
- Processes, checks, updates
- An invoice or stock agent

THE TIEBREAKER

Someone is chatting with it: conversational. It wakes up when something happens and nobody chats: a background workflow agent.

Deterministic, generative or hybrid?

KIND	BEST FOR
Deterministic	Fixed steps, exact wording, full audit
Generative	Open requests and natural conversation
Hybrid	Fixed where it must be, flexible elsewhere

ON THE EXAM

The guide stresses all three. Regulated or irreversible steps lean deterministic; open conversation leans generative.

Six kinds of agent, by the work they do

CATEGORY	AT WRENFIELD
Customer agents	Shopping help, returns, order questions
Employee agents	Policy answers, rota drafts, IT requests
Creative agents	Campaign copy, images and video
Data agents	Questions answered from sales data
Code agents	Write, test and explain code
Security agents	Spot and respond to threats

ON THE EXAM

A question may describe one and ask which category it is. Match who it serves and what it produces.

Multi-agent systems: a kitchen brigade

- **An orchestrator** splits the goal and hands out the parts.
- **Specialist agents** each do one job well, with their own tools.
- **Each can use a different model,** sized to its task.
- **Smaller parts** are easier to build, test, govern and replace.
- **Agents talk** through standard protocols such as A2A, in module 18.

> THE SIGNAL
>
> Several distinct skills, tested and governed separately: a multi-agent system with an orchestrator.

One agent or a team?

SINGLE AGENT

One well-defined job

- A clear, bounded task
- One model, a few tools
- Simpler to run and debug
- Lower cost

MULTI-AGENT

Many distinct skills

- Several specialised sub-tasks
- Different models and tools
- Each part governed alone
- More coordination to manage

THE TIEBREAKER

Start with one agent. Split into a team only when the skills, tools or controls genuinely differ.

More autonomy needs more guardrails

- **Least privilege:** give each agent only the tools and data its job needs.
- **Human approval** for high-impact or irreversible actions.
- **Log every action,** so decisions can be traced and audited.
- **Test before launch,** including unusual and hostile inputs.
- **Start narrow,** then widen autonomy as trust is earned.

> REMEMBER
>
> The more an agent can do on its own, the more the controls matter. Modules 26 and 27 go deeper.

When an agent is the wrong tool

SITUATION	BETTER CHOICE
One question, one answer	A prompt, or the Gemini app
Fixed steps, no decisions	Rules or a simple workflow
Deep empathy, such as a bereavement call	A person
High ethical stakes: credit, hiring	People decide; AI supports
A small job on a tight budget	Something simpler and cheaper

THE TAKEAWAY

An agent adds cost, risk and complexity. Use one only when the job needs reasoning and action over several steps.

> Wrenfield's website helper recommends three sofas, and the customer then orders one.
>
> The new version takes a goal, such as "furnish my living room under 1,500 pounds". It picks items, checks stock, applies the loyalty voucher, and places the order once the customer confirms.

What best describes the new version?

A A rules-based bot

B An AI assistant

C An AI agent

D A search engine

◆ PAUSE AND DECIDE

What best describes the new version?

A A rules-based bot

Bots follow set rules; this one plans and acts.

B An AI assistant

The old helper. Assistants recommend; the user acts.

C An AI agent

Goal-driven, multi-step, using tools to act.

D A search engine

Search finds items; it does not complete a goal.

THE SIGNAL

Given a goal, it plans and acts over several steps: an agent.

> Wrenfield's returns agent tries to book a courier collection for tomorrow.
>
> The courier's system replies: no slots are available tomorrow.

What should a well-designed agent do next?

A Check the result, re-plan with the next free slot, and confirm with the customer

B Tell the customer the collection is booked anyway

C Cancel the whole return

D Retry the same booking until it works

◆ PAUSE AND DECIDE

What should a well-designed agent do next?

A Check the result, re-plan with the next free slot, and confirm with the customer

Observe, re-plan, act: the loop working as intended.

B Tell the customer the collection is booked anyway

A false confirmation. The worst outcome.

C Cancel the whole return

Gives up on a problem it could solve.

D Retry the same booking until it works

Ignores the result; it may never succeed.

THE SIGNAL

Check the result, then loop again: observe, reason, plan, act.

> About 52,000 supplier invoices and delivery notes arrive by email each month, at all hours.
>
> Wrenfield wants each one read, checked against the purchase order, and posted to the finance system, with nobody chatting to the system.

Which type of agent fits?

A. A conversational agent in the customer chat

B. A background agent, started by each new invoice

C. Each finance clerk using the Gemini app

D. A Gem with invoice instructions

◆ PAUSE AND DECIDE

Which type of agent fits?

A A conversational agent in the customer chat

Talks with customers; nobody is chatting here.

B A background agent, started by each new invoice

Event-driven, behind the scenes, no conversation.

C Each finance clerk using the Gemini app

Still a person on every invoice.

D A Gem with invoice instructions

Saved instructions; it cannot act on its own.

THE SIGNAL

Started by an event, with nobody chatting: a background agent.

> Wrenfield's returns agent understands requests well and makes sensible plans.
>
> But it cannot actually check an order's status or book a courier, so it only tells customers what should happen.

What is the agent missing?

A. A larger, more capable model

B. More memory

C. Longer instructions

D. Tools connected to the orders system and courier

◆ PAUSE AND DECIDE

What is the agent missing?

A A larger, more capable model

The brain is already fine.

B More memory

Memory recalls; it does not act.

C Longer instructions

Instructions guide; they cannot reach systems.

D Tools connected to the orders system and courier

Tools are the hands that let it act.

THE SIGNAL

It thinks well but cannot act: it needs tools.

> Returning customers complain that Wrenfield's shopping agent forgets them.
>
> It never recalls their room sizes, colour preferences or past returns from earlier visits.

Which part should the team add or improve?

A Long-term memory

B A bigger context window

C A lower temperature

D More tools

◆ PAUSE AND DECIDE

Which part should the team add or improve?

A **Long-term memory**
Keeps what it learned across visits.

B **A bigger context window**
Holds more within one session, not across visits.

C **A lower temperature**
Steadier wording; it does not remember anything.

D **More tools**
More actions; still nothing remembered.

THE SIGNAL

Forgets across visits: long-term memory.

> Wrenfield's Netherlands launch needs 48,000 product descriptions translated, checked against Dutch labelling rules, and loaded into the catalogue.
>
> Each step needs different skills and tools, and the team wants to test and govern each step separately.

Which design fits best?

A One large agent with every tool

B The Gemini app, used by each copywriter

C An orchestrator agent coordinating specialist agents

D A Gem for each step

◆ PAUSE AND DECIDE

Which design fits best?

A One large agent with every tool

Hard to test, govern or fix one part.

B The Gemini app, used by each copywriter

Manual, and far too slow for 48,000 items.

C An orchestrator agent coordinating specialist agents

Each specialist tested and governed on its own.

D A Gem for each step

Saved instructions cannot coordinate or act.

THE SIGNAL

Distinct skills, governed separately: a multi-agent system.

> Wrenfield's HR team proposes an agent that decides which of 30,000 seasonal applicants to reject, and emails them.
>
> Rejected applicants from the last pilot are already asking why.

What should a leader recommend?

A Let a fully autonomous agent decide and send the emails

B An agent summarises applications, and people decide and record reasons

C A team of agents that ranks and rejects applicants

D A deterministic flow that rejects on keywords

◆ PAUSE AND DECIDE

What should a leader recommend?

A Let a fully autonomous agent decide and send the emails

High ethical stakes, with no human judgement.

B An agent summarises applications, and people decide and record reasons

AI supports; people decide, and reasons are kept.

C A team of agents that ranks and rejects applicants

More agents, same problem: nobody accountable.

D A deterministic flow that rejects on keywords

Rigid, unfair, and still unexplained.

THE TAKEAWAY

High ethical stakes: people decide, and AI supports them.

> Wrenfield's returns agent will be allowed to issue refunds.
>
> Leadership worries about large or mistaken refunds going out unnoticed.

What is the best control?

A Limit its permissions, require human approval above a set amount, and log every action

B Trust it, because the model is highly accurate

C Turn the agent off

D Add "please be careful" to its instructions

◆ PAUSE AND DECIDE

What is the best control?

A Limit its permissions, require human approval above a set amount, and log every action

Least privilege, approval and audit, together.

B Trust it, because the model is highly accurate

Accuracy is not a control.

C Turn the agent off

Loses the benefit entirely.

D Add "please be careful" to its instructions

Instructions are not a guarantee.

THE TAKEAWAY

Autonomy with guardrails: limits, approval and logs.

Module 16 in five lines

- A bot follows rules, an assistant recommends, an agent pursues a goal and acts.
- The loop: observe, reason, plan, act, check, and go round again.
- Inside: model, instructions, tools, memory and orchestration.
- Conversational or background; deterministic, generative or hybrid; one agent or a team.
- Add guardrails as autonomy grows, and skip agents when simpler tools will do.